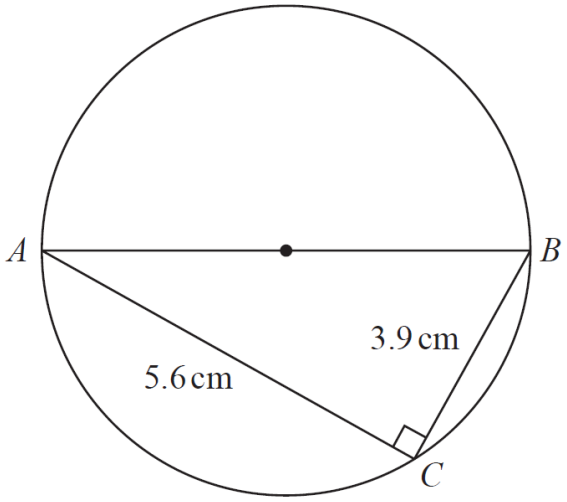
27 A, B and C are points on a circle with diameter AB.
ABC is a right-angled triangle.



Calculate the area of the circle.
Give your answer correct to 3 significant figures.

$AB^2 = 5.6^2 + 3.9^2 = 46.57$

1 mark
Use Pythagoras.

$AB = \sqrt{46.57} = 6.824221567...$ cm

1 mark
Find the diameter.

Radius $= AB \div 2 = 3.412110783...$ cm

Area $= \pi r^2$

1 mark
Halve diameter, then use πr^2.

$= \pi \times (\sqrt{46.57} \div 2)^2$

$= 36.57599247...$ cm^2

Area $= 36.6$ cm^2 (3 s.f.)

1 mark
36.6 cm^2.

…….. cm^2

(Total for Question 27 is 4 marks)